MODULE 09 · SECTION 1 - FUNDAMENTALS OF GEN AI

Choosing a model, and when not to use gen AI

Seven factors for picking a model, why bigger is not better, and the jobs where generative AI is the wrong tool.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 9 of 33

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Start from the job, not the model. And sometimes the best model is **no generative model at all**.

Seven factors for choosing a model

Modality
What goes in
What comes out

Context window
How much it reads
in one request

Security
Where data goes
Who can see it

Reliability
Consistent quality
Always available

Choosing a model
Seven factors
from the exam guide

Describe the job, then test each factor

Cost
Price per request
At your volume

Performance
Quality of output
Speed of reply

Customisation
How far you
can adapt it

ON THE EXAM

Questions describe one hard constraint, then ask which factor decides. Find the constraint first.

Modality: name both directions

- **Inputs:** text, images, audio, video, code, documents.
- **Outputs:** text, code, images, video, speech.
- **Direction matters:** photo in, text out is not text in, photo out.
- **Check it first:** it rules a model in or out before anything else.
- **Multimodal** models such as Gemini take many inputs in one request.

> THE TIEBREAKER
>
> Write the job as "X in, Y out". The model must handle both ends.

Context window: the size of the desk

- **The desk:** everything the model can see at once.
- **It holds** your instructions, documents, chat history and the reply.
- **Measured in tokens;** many Gemini models take 1 million or more.
- **That fits** long contracts, hours of audio, or a whole codebase.
- **A bigger desk** costs more per request and replies more slowly.

> THE TIEBREAKER
>
> Long material read in one go: context window. Facts the model never learned: grounding, in module 21.

Security and reliability

SECURITY

Who can see the data?

- Enterprise terms on data use
- Choice of processing region
- Access controls and permissions
- Safety filters on inputs and outputs

RELIABILITY

Will it behave every time?

- Consistent quality on your task
- Available when customers need it
- Stable, versioned models
- Answers you can check

THE SIGNAL

Customer or staff data in the story: security decides. A live, customer-facing service: reliability decides.

Bigger is not better. You don't send a lorry to deliver a postcard.

When the smaller model wins

SITUATION	WHY SMALLER WINS
Millions of simple requests	Cost per request adds up fast
A shopper is waiting	Speed matters more than depth
A short, routine task	Extra reasoning adds nothing
It must run on a device	It has to fit the hardware

THE TIEBREAKER

Try the smallest model that meets the quality bar. Move up a tier only when it falls short.

Customisation: climb only as far as you need

01
Prompt
Instructions and examples. Fastest and cheapest.

›

02
Ground
Connect your own data when it answers.

›

03
Tune
Adapt a model with your own examples.

›

04
Train
Build a new model. Rarely worth it.

ON THE EXAM

Training a new foundation model is almost never the answer for a business. Start at the bottom of the ladder.

Open models or managed models?

OPEN

Weights in your hands

- Gemma, and partner models
- Download and run anywhere
- Inspect and tune freely
- You run, patch and scale it

PROPRIETARY

Managed for you

- Gemini, on Agent Platform
- Most capable models
- Google runs and scales it
- Pay per use, no servers

REMEMBER

Open does not mean free. The licence may cost nothing, but servers and people do.

Generative models are probabilistic. When the answer must be **exact, repeatable and explainable**, look elsewhere first.

Generative AI, traditional ML or plain rules?

TOOL	ITS JOB
Generative AI	Create, converse, summarise, extract
Traditional ML	Classify, predict, forecast, cluster
Rules and software	Fixed calculations and lookups

CURRENCY CHECK

Google's guide to choosing generative or traditional AI, updated September 2026, draws the same line.

Signs that gen AI is the wrong tool

- The same input must give **the same output** every time.
- Each decision needs its **real reasons** on record, for an auditor.
- The answer **already sits in a database**.
- The task is a **calculation or a fixed rule**.
- It is a **prediction from structured history**: traditional ML.

> ON THE EXAM
>
> An option that uses gen AI where a rule, a lookup or a prediction model would do is usually the distractor.

The Wrenfield Pay trap

- **The proposal:** Gemini reads each credit application and decides.
- **Problem one:** the same applicant could get a different answer.
- **Problem two:** a written explanation is not the real cause.
- **Problem three:** a regulated lender must audit every decision.
- **Better:** a credit model or rules decide; Gemini drafts the letter.

> THE TAKEAWAY
>
> Put gen AI around the decision, not in charge of it.

Often the answer is both

- **Google's guide:** traditional and generative AI aren't mutually exclusive.
- **Segment, then write:** clustering groups customers; gen AI writes for each.
- **Predict, then talk:** a churn model flags risk; a chatbot reaches out.
- **Forecast, then explain:** demand figures, plus a plain-English summary.
- **Decide, then draft:** the credit pattern from the last slide.

> THE TIEBREAKER
>
> If one option combines the right tool for each step, it often beats a pure gen AI option.

Six business applications of gen AI

APPLICATION	AT WRENFIELD
Text generation	Product copy and reply drafts
Image generation	Campaign and lifestyle images
Code generation	Tests and boilerplate for developers
Video generation	Short social media adverts
Data analysis	Themes in reviews and call notes
Personalisation	Emails tailored to each shopper

ON THE EXAM

A question may describe one of these and ask you to name it. Match the output, not the department.

> Wrenfield's warehouse films each damaged return on arrival. The quality team wants a written damage report for every clip.
>
> They are shortlisting models and want to rule out unsuitable ones quickly.

Which factor should rule models in or out first?

A Context window

B Modality

C Cost

D Customisation

◆ PAUSE AND DECIDE

Which factor should rule models in or out first?

A Context window

Matters for long clips, but only once a model takes video at all.

B Modality

Video in, text out. A model that cannot read video is out at once.

C Cost

Worth comparing later, among models that can do the job.

D Customisation

Adapting comes after you have a model that fits the job.

THE SIGNAL

Video in, text out: modality first.

> Wrenfield's legal team holds 40 supplier contracts of about 300 pages each.
>
> They want a model to read a full set of contracts together and flag clauses that conflict across them.

Which factor matters most here?

A Modality

B Security

C Context window

D Performance

◆ PAUSE AND DECIDE

Which factor matters most here?

A **Modality**

Text in, text out. Almost any model handles that.

B **Security**

Worth checking, but nothing in the story turns on it.

C **Context window**

Many long contracts read together must fit in one request.

D **Performance**

Speed is not the constraint. Nobody is waiting at a till.

THE SIGNAL

Lots of long material read together: context window.

> Wrenfield's HR team wants an assistant that answers questions about staff records.
>
> The chief information security officer insists the data stays in a UK region, is never used to train a provider's models, and is only seen by authorised staff.

Which factor dominates the choice?

A Security

B Reliability

C Context window

D Customisation

◆ PAUSE AND DECIDE

Which factor dominates the choice?

A Security

Region, data use and access controls are all security.

B Reliability

Matters for any service, but the demands here are about data.

C Context window

Nothing suggests the records are too long to fit.

D Customisation

Nothing here asks to adapt the model's style or task.

THE SIGNAL

Where data goes and who can see it: security.

> Wrenfield wants search suggestions to appear as shoppers type, on millions of searches a day.
>
> Each suggestion is a short, simple completion, and it must appear almost instantly.

Which choice fits best?

A The largest Pro model, with thinking switched on

B Train a new foundation model on Wrenfield's search logs

C Gemma, hosted on Wrenfield's own GPUs

D A small, fast tier such as Gemini Flash-Lite

◆ PAUSE AND DECIDE

Which choice fits best?

A The largest Pro model, with thinking switched on

Most capable, but slower and costlier than this job needs.

B Train a new foundation model on Wrenfield's search logs

Vast cost and effort for a short completion.

C Gemma, hosted on Wrenfield's own GPUs

Possible, but adds servers to run for no stated reason.

D A small, fast tier such as Gemini Flash-Lite

Simple, high-volume and instant: the small, fast tier fits.

THE SIGNAL

Simple, high volume, instant: small and fast beats big.

> A manager says open models are free, so Wrenfield should host one to draft customer replies.
>
> Wrenfield has nobody to run model servers, and wants the best reply quality it can get.

What should a leader recommend?

A. Host Gemma on new GPUs, since the licence costs nothing

B. Train a new model so Wrenfield owns it outright

C. Use Gemini, managed on Agent Platform, paid per use

D. Download the largest open model available and run it

◆ PAUSE AND DECIDE

What should a leader recommend?

A Host Gemma on new GPUs, since the licence costs nothing

Free licence, but servers and staff to run them are not.

B Train a new model so Wrenfield owns it outright

The top rung of the ladder. Far too costly for drafting.

C Use Gemini, managed on Agent Platform, paid per use

Most capable, and Google runs it. Nothing to host.

D Download the largest open model available and run it

Bigger and still self-run. The team cannot run it.

THE SIGNAL

Open does not mean free. No team to run it: choose managed.

> Wrenfield wants to forecast weekly demand for each product in each store.
>
> It holds five years of sales history in BigQuery, as rows of structured data.

Which approach fits best?

A Paste the sales table into Gemini and ask it to predict

B Train a forecasting model on the sales history

C Write a fixed rule that reorders at a set stock level

D Use Gemini Notebook to summarise past sales reports

◆ PAUSE AND DECIDE

Which approach fits best?

A Paste the sales table into Gemini and ask it to predict

Gen AI is not built to forecast from rows of numbers.

B Train a forecasting model on the sales history

Prediction from structured history is traditional ML.

C Write a fixed rule that reorders at a set stock level

A fixed rule ignores trends, seasons and promotions.

D Use Gemini Notebook to summarise past sales reports

Summarises the past. It does not forecast the future.

THE SIGNAL

A prediction from structured history: traditional ML, not gen AI.

> Wrenfield Pay must give every declined applicant the main reasons for the decision.
>
> A director proposes that Gemini should read each application and decide.

What should a leader recommend?

A Let Gemini decide, with temperature set to zero

B Fine-tune Gemini on past credit decisions

C Let Gemini decide and explain its own reasoning

D A credit model decides, and Gemini drafts the letter

◆ PAUSE AND DECIDE

What should a leader recommend?

A Let Gemini decide, with temperature set to zero

Less random, but still not an auditable decision.

B Fine-tune Gemini on past credit decisions

Better tuned, but still probabilistic and hard to audit.

C Let Gemini decide and explain its own reasoning

A fluent explanation is not proof of the real cause.

D A credit model decides, and Gemini drafts the letter

Repeatable, recorded reasons; gen AI only writes the letter.

THE TAKEAWAY

Exact, repeatable, auditable decisions: gen AI around the decision, not in charge.

> Wrenfield wants a monthly email for its 410,000 cardholders.
>
> First, customers should be grouped by buying behaviour. Then each group should get copy written for its interests.

Which approach fits best?

A Clustering to form the groups, then gen AI to write each email

B Gen AI to invent the groups and write every email

C Traditional ML for both the groups and the copy

D Rules to form the groups, then a human writes every email

◆ PAUSE AND DECIDE

Which approach fits best?

A Clustering to form the groups, then gen AI to write each email

Right tool for each step: clustering, then generation.

B Gen AI to invent the groups and write every email

Grouping customers from data is a clustering job.

C Traditional ML for both the groups and the copy

Traditional ML does not write fresh, tailored copy.

D Rules to form the groups, then a human writes every email

Rigid groups, and far too slow for many segments.

THE SIGNAL

Two steps, two tools: hybrid beats pure gen AI.

Module 9 in five lines

- Seven factors: modality, context window, security, reliability, cost, performance, customisation.
- Write the job as "X in, Y out", then find the one hard constraint.
- Bigger is not better: try the smallest model that meets the bar.
- Open does not mean free; managed means nothing to run.
- Exact, repeatable, auditable? Rules or traditional ML, with gen AI around it.